

SolarKits v2.0

Project Development Status & Feature Documentation Report

Audit Date: August 29, 2026

Document Type: Complete Project Audit & Technical Architecture Documentation

Payment Architecture: Offline Bank Transfer + Payment Receipt Upload & Verification

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Mode: Read-Only Static Code Analysis — No Application Source Code Modified

Classification: CONFIDENTIAL — Internal & Management Use Only

Applications	Backend Modules	Features Documented	Bugs / Risks	Code Completion
7	9	92	12	77.0%

■ CRITICAL SECURITY NOTE: CORS bypass and KYC/receipt privacy must be addressed — see Section 8

1. Executive Summary

SolarKits v2.0 is an enterprise-grade, multi-tenant B2B Solar E-Commerce, Channel Distribution, and Operations Ecosystem. The platform integrates B2B EPC contractors, regional franchise partners, BOS distributors and dealers, warehouse logistics, financial accounts, and central administrators into a unified digital platform powered by a Node.js/Express 5 backend and React 19 frontends.

Payment Architecture (Offline Model): The platform operates on a **100% Offline Bank Transfer & Payment Receipt Upload Workflow**. Franchise partners, EPCs, and distributors transfer funds offline (via NEFT, RTGS, IMPS, or Cheque) directly into the company's bank account and upload the bank payment slip along with the UTR number into the portal (via POST /api/india/v1/reseller/fee-payment/upload-receipt). The Admin / Accounts team then reviews the slip, validates the UTR against company bank statements, and activates franchise accounts / approves orders. Commission payouts and net Franchise Earning are settled manually via direct bank transfer with UTR logging.

Code-Audit-Based Overall Completion Estimate: 77.0%

Feature Status Distribution:

Development Status	Feature Count	Weight Factor
Completed but Not Functionally Verified	74	0.85
Frontend Only	3	0.35
Not Started	2	0.00
Partially Completed	7	0.50
Static or Mock Implementation	1	0.20
Completed and Verified	3	1.00
Broken or Suspected Broken	2	0.10

2. Applications & Technology Overview

Application	Source Folder	Port	Technology Stack	Status
Central Backend API	backend/solarkits-central-backend	5000	Node.js/Express 5, Mongoose 9, MongoDB	Partially Completed
Unified Internal Admin Portal	internal-admin-portal/solarkits-unified-admin	5174	React 19, Vite 7, Redux Toolkit, Tailwind v4	Partially Completed
Franchise Portal	customer-apps/solarkits-reseller-portal	5178	React 19, React Router v7, Tailwind v4	Partially Completed
Direct EPC Solar Store	customer-apps/solar-store	5177	React 19, Redux Toolkit, Tailwind v3	Partially Completed
SolarShop India Marketplace	customer-apps/solarkits-solarshop-india	5173	React 19, Redux, Tailwind v4	Static or Mock Implementation
BOSKIT B2B Platform	customer-apps/boskit-website	5180	React 19, Tailwind v4	Partially Completed
BDE Field Agent Portal	customer-apps/BDE-modules	N/A	React 19, Vite, Tailwind	Partially Completed

2.1 Live Deployment Information

Per README.md, three applications are deployed on Render.com: (1) solarkits-unified-admin.onrender.com, (2) reseller-portal.onrender.com (Franchise Portal), (3) solarshop-india.onrender.com. Docker multi-stage build and Nginx templates are configured for production containerization.

3. Technology Stack Reference

Layer	Technology	Version / Architectural Role
Backend Runtime	Node.js	v18+ / v22 LTS
Backend Framework	Express.js	v5.x (async error handling built-in)
Database ORM	Mongoose	v9.x connecting to MongoDB
Frontend Framework	React	v19.x across all portals
Build Tool	Vite	v7.x
State Management	Redux Toolkit	Configured across all customer & admin portals
CSS Framework	Tailwind CSS	v3.4 (Solar Store) / v4.1 (others)
Animations	Framer Motion	Latest
Payment Architecture	Offline Bank Transfer	Receipt Slip Upload + UTR Tracking + Admin/Accounts Verification
Media / File Storage	Cloudinary	v2.x SDK — used for KYC & Payment Receipt slip uploads
Email	Nodemailer	v9.x — SMTP/Gmail configured
SMS / WhatsApp	Twilio	v6.x — OTP delivery
PDF Generation (backend)	pdfkit	v0.19.1
GST Verification	QuickEKYC	Sandbox and production modes
Authentication	JWT (RS256 + HMAC)	Access: 15min / Refresh: 2 days / httpOnly cookies
Password Hashing	bcrypt	v6.x
Containerization	Docker + Nginx	Multi-stage build + reverse proxy
Deployment	Render.com	3 apps live on onrender.com
Database Architecture	MongoDB Multi-Database	7 logical DBs: user, core, india_solarshop, boskit, geolocation, warehouse, supplier

4. Database Architecture & Entity Inventory

Database Name	Schemas	Key Entities & Collections	Primary Consumers
user_db	17 schemas	cms_users, cms_roles, cms_modules, cms_panels, otps, saas_products, user_panels	Admin, Warehouse, Accounts, Ops, Dev, BDE portals
core_db	54 schemas	products, product_templates, solar_kits, combo_kits, combo_kit_variants, industry_types, project_categories, brands, warehouses, po_settings, commission_rules, moq_rules, website_content, industry_content, geolocation	Admin, Solar Store, Franchise Portal
india_solarshop_db	19 schemas	epc_accounts, bos_kits, cart, offer_masters, request_orders, solarshop_settings, inventory_reservations, reseller_wallets, reseller_ledger, reseller_plan_subscriptions	Solar Store, Franchise Portal, BDE
boskit_db	27 schemas	boskit_distributors, boskit_dealers, boskit_distributor_plans, boskit_orders, boskit_payments, boskit_commissions, boskit_notifications, boskit_territories, boskit_moq_rules, boskit_tax_rules, boskit_channel_settings, boskit_plan_versions	BOSKIT Platform
company_warehouse_db	N/A	Company warehouses, stock inward logs, SKU serials, inventory allocations, purchase orders, dispatch records	Warehouse Panel, Accounts Panel
geolocation_db	N/A	Countries, states, districts, pincodes, geo boundaries	All territory-aware modules
supplier_db	N/A	Suppliers, brands, supplier contracts, bid records	Supplier Panel, Accounts Panel

5. User Roles, Panels & Access Architecture

Role	Panel / Portal	Authentication	Key Responsibilities & Scope
Super Admin	Admin Portal (/admin-panel/*)	CMS Auth OTP/Email	Full governance, master catalogs, user creation, receipt approval, territory, commissions
Accounts Team	Accounts Portal (/account-panel/*)	CMS Auth OTP/Email	Receipt verification, GST invoicing, PO creation, supplier management, Franchise Earning payouts, UTR tracking
Warehouse Staff	Warehouse Portal (/warehouse/*)	CMS Auth OTP/Email	Stock inward, outward, delivery, vehicle/driver management
Operations Team	Operations Portal (/operation-management-panel/*)	CMS Auth	Order dispatch, tracking, project milestone management
Developer	Developer Portal (/developer-panel/*)	CMS Auth	Module activation toggles, system health diagnostics
BDE Field Agent	BDE Portal (standalone app)	Email/Password JWT	Lead creation, franchise pipeline, store setup verification, goal tracking
Franchise Partner	Franchise Portal (standalone app)	Mobile OTP/PIN	KYC, offline receipt upload, product catalog, margin, storefront, EPC management, Franchise Earning tracking
BOSKIT Distributor	BOSKIT Website + Admin	BOSKIT Auth	Plan purchase, territory, procurement, dealer management, catalogue
BOSKIT Dealer	BOSKIT Website	BOSKIT Auth	Catalogue, cart, checkout, order tracking
EPC Contractor	Solar Store (standalone app)	Email/Password JWT	Kit browsing, custom config, bulk buy, cart, order confirmation, order tracking
BOSKIT Admin	Admin Portal (/boskit-admin/*)	CMS Auth	Distributor applications, plan management, territory, MOQ/GST rules, reports
Supplier	Supplier Panel (API)	Supplier Auth	Brand catalog, bid submissions, supply agreements

6. Feature Scope of Work & Functional Matrix (All 92 Features)

Detailed functional scope of work and business process description for each platform feature:

ID	Module	Feature Name	Scope of Work & Functional Description	Integration
F001	Authentication	CMS Multi-Role Login	Central unified authentication for Super Admin, Accounts, Warehouse, Operations, and Developers. Validates email/password + OTP, checks panel permissions, issues RS256 JWT access (15m) & refresh (2d) tokens via httpOnly cookies, and redirects users to their assigned department portal.	Fully Connected
F002	Authentication	BDE Login	Dedicated authentication service for Business Development Executives. Validates credentials, attaches assigned geographic territory and franchisee pipeline scope to the user session.	Fully Connected
F003	Authentication	Franchise Login & Registration	Mobile number OTP verification and 4-digit PIN setup for regional Franchise Partners. Handles session token management, account status verification, and rate limiting against brute-force attacks.	Fully Connected
F004	Authentication	EPC Contractor Login/Signup	Self-service onboarding and authentication for EPC solar contractors. Grants access to preconfigured combo kits, custom kit builders, and bulk purchasing with contractor pricing.	Fully Connected
F005	Authentication	BOSKIT Distributor Auth	Multi-tier authentication portal for BOS distributors and local electrical dealers with role-based permission scopes, territory binding, and dealer network association.	Fully Connected
F006	BDE Management	Admin Creates BDE	Super Admin interface to onboard new BDE field agents, upload identity KYC documents (Aadhaar, PAN), assign territorial states/districts, and configure monthly franchisee onboarding targets.	Fully Connected
F007	BDE Management	BDE Dashboard	Operational command dashboard for field agents displaying real-time lead counts, pipeline conversion velocity, store setup milestone progress, and monthly target fulfillment metrics.	Fully Connected
F008	BDE Management	BDE Lead Management	Comprehensive lead tracking module allowing BDEs to register prospective franchise leads, schedule customer visits, record meeting minutes, log activity trails, and advance leads through onboarding stages.	Fully Connected
F009	BDE Management	BDE Store Setup (16-Step)	16-step physical store setup verification engine (branding, signage, display kits, IT infrastructure). BDE uploads on-site photo proofs, submits delay exception requests, and requests final admin certification.	Fully Connected
F010	BDE Management	BDE Franchise Pipeline	Pipeline tracking visualization monitoring converted leads through KYC verification, offline fee receipt upload, physical store setup, and final operational activation.	Fully Connected
F011	BDE Management	Admin BDE Dashboard	Admin oversight dashboard monitoring field agent KPIs, tracking lead-to-onboarding conversion rates, reviewing KYC submissions, and dynamically reallocating territory quotas.	Fully Connected
F012	BDE Management	BDE Territory Assignment	Administrative tool to map geographic boundaries (states, districts, clusters) to specific BDE agents with strict territory exclusivity enforcement.	Fully Connected
F013	BDE Management	BDE Activity History	Complete audit trail logging BDE field interactions, status changes, photo proof uploads, and franchisee verification submissions with timestamps.	Fully Connected
F014	Franchise Module	Franchise Landing Page	Public portal for prospective franchise partners showcasing plan benefits, ROI projections, investment tiers, and an interactive district territory availability checker.	Fully Connected
F015	Franchise Module	KYC Workspace (GST, PAN, Aadhaar)	Structured verification workspace where applicants upload GST certificates, PAN cards, and Aadhaar documents. Features live GST verification via QuickeKYC and secure document storage.	Fully Connected
F016	Franchise Module	Offline Fee Payment & Receipt Upload	Applicants execute offline bank transfer (NEFT/RTGS/IMPS/Cheque) for franchise plan fees, upload the bank payment slip, and enter the UTR number. Updates status to 'receipt_uploaded' for admin approval.	Fully Connected

F017	Franchise Module	Franchise Dashboard	Operational homepage for active franchise partners displaying real-time Franchise Earning totals, commission summaries, EPC order pipeline, assigned territory leads, and recent dispatch milestones.	Fully Connected
F018	Franchise Module	Authorized Product Catalog	Catalog allowing franchise partners to browse admin-approved solar kits, configure regional profit margins, and selectively publish systems to local EPC contractors.	Fully Connected
F019	Franchise Module	Franchise Earning & Commission	Digital earnings ledger tracking real-time commission accruals and net Franchise Earning from EPC orders. Enables franchise partners to monitor balances and submit bank withdrawal requests for manual payout processing.	Fully Connected
F020	Franchise Module	EPC Buyer Management	CRM module enabling franchise partners to register local EPC contractors, manage customer relationships, and track regional contractor purchase volume.	Fully Connected
F021	Franchise Module	Franchise PO Orders	Purchase order engine allowing franchise partners to procure bulk stock from central warehouses based on tier-specific discount rules and credit limits.	Fully Connected
F022	Franchise Module	Store Setup Tracking	Franchisee-facing progress tracker displaying status of the 16 physical store setup milestones and assigned BDE agent inspection schedules.	Fully Connected
F023	Franchise Module	Procurement & Inventory	Franchise inventory management portal to order bulk solar equipment, log incoming warehouse deliveries, and maintain local stock levels.	Fully Connected
F024	Franchise Module	Storefront Listings (Publish to EPC)	Pricing and listing manager enabling franchise partners to adjust selling prices, apply percentage/fixed margins, and publish products to EPC storefronts.	Fully Connected
F025	Franchise Module	Territory Eligibility Checker	Interactive district search allowing prospective applicants to check territory exclusivity availability before submitting franchise applications.	Fully Connected
F026	Franchise Module	PlansPortal (Franchise Plans)	Showcase portal displaying franchise package tiers, upfront fee schedules, commission rates, and territory rights (UI present; route currently commented in App.jsx).	Frontend Only
F027	Franchise Module	Loose Order	Component procurement module for purchasing standalone non-bundled solar parts (panels, inverters, cables) outside standard combo kits (currently a placeholder stub).	Not Started
F028	Admin Panel	Product Template Management	Master catalog engine to create standardized product templates with configurable technical attribute groups, specifications, and automated SKU generation.	Fully Connected
F029	Admin Panel	Solar Kits Management	Admin module to bundle individual solar components into complete solar kits (On-Grid, Off-Grid, Hybrid), configuring baseline pricing and warehouse activations.	Fully Connected
F030	Admin Panel	Combo Kits & Variants	Advanced Bill of Materials (BOM) configurator allowing creation of multi-tier combo kits with interchangeable inverter brands, panel ratings, and structure options.	Fully Connected
F031	Admin Panel	Industry Type & Project Hierarchy	Taxonomy hierarchy manager structuring solar systems across Industry > Category > SubCategory > SystemType > Capacity tiers.	Fully Connected
F032	Admin Panel	Industry Content CMS	Dynamic marketing CMS to upload promotional banners, video testimonials, YouTube walkthroughs, and technical posters categorized by industry segment.	Fully Connected
F033	Admin Panel	Website Content CMS	Content management engine for public websites controlling hero banners, FAQs, terms of service, partner testimonials, and promotional announcements.	Fully Connected
F034	Admin Panel	Geolocation Management	Geographic master database managing countries, states, districts, pin codes, and territory cluster boundaries across India.	Fully Connected
F035	Admin Panel	User & Role Management	RBAC security management engine defining internal departments, panel permissions, module access rights, and staff account creation.	Fully Connected

F036	Admin Panel	Franchise Management & Receipt Approval	Central verification interface to review franchise KYC submissions, inspect uploaded payment receipt slips, verify bank UTRs, track Franchise Earning, and activate operational accounts.	Fully Connected
F037	Admin Panel	Store Setup Administration	Admin governance tool to inspect 16-step BDE store setup photo proofs, approve delay exception requests, and certify new franchise store openings.	Fully Connected
F038	Admin Panel	Franchise PO Settings	Configuration manager establishing minimum order quantities (MOQ), payment terms, and credit thresholds for franchise purchase orders.	Fully Connected
F039	Admin Panel	Commission & MOQ Rules	Rules engine configuring percentage and fixed commission formulas for franchise partners based on sales volumes and kit categories.	Fully Connected
F040	Admin Panel	Warehouse Management (Admin)	Master warehouse setup managing physical warehouse locations, stock thresholds, SKU activations, and regional dispatch radius.	Fully Connected
F041	Admin Panel	Supplier Management	Supplier registry managing solar equipment manufacturers, brand partnerships, product supply lines, and contract pricing.	Fully Connected
F042	Admin Panel	EPC Management	Contractor registry monitoring registered EPC installers, account statuses, and territorial activity.	Partially Connected
F043	Admin Panel	Approve New EPC	Verification workflow to review and approve newly registered EPC contractor accounts.	Partially Connected
F044	Admin Panel	Price Master (SKU Pricing)	Centralized pricing matrix controlling base SKU costs, benchmark rates, and dynamic price approval workflows.	Fully Connected
F045	Admin Panel	Units Management	Measurement configuration managing packaging units, weights, electrical metrics (kW, Wp, Ah, V), and conversion factors.	Fully Connected
F046	Solar Store (EPC)	Preconfigured Combo Kits	EPC store catalog displaying turnkey solar kits with real-time district warehouse inventory polling every 30 seconds.	Fully Connected
F047	Solar Store (EPC)	Custom Combo Kit Builder	Interactive kit builder enabling EPC contractors to customize panel brands, inverter capacities, and mounting structures with dynamic BOM pricing.	Fully Connected
F048	Solar Store (EPC)	Bulk Buy	Volume procurement portal offering tiered bulk discounts for commercial and industrial solar installations.	Fully Connected
F049	Solar Store (EPC)	Cart & Order Confirmation (Offline Receipt / Request)	Cart and checkout engine featuring 30-minute stock reservation, GST validation, and offline bank transfer receipt attachment.	Fully Connected
F050	Solar Store (EPC)	EPC Catalogue (Tenant-Isolated)	Tenant-isolated catalog ensuring EPC contractors view regional franchise pricing while manufacturer baseline costs remain completely hidden.	Fully Connected
F051	Solar Store (EPC)	Store Locator	Geographic locator interface displaying nearby company warehouses and franchise stores for self-pickup of solar equipment.	Fully Connected
F052	Solar Store (EPC)	Request Order	Custom BOM request tool for EPC contractors to submit non-standard project bills of materials for offline quotation (route mapping pending in Board.jsx).	Frontend Only
F053	Solar Store (EPC)	Solar BOS Kit (Commented Out)	Specialized catalog for Balance of System electrical accessories (DCDB, ACDB, cables, earthing kits) — backend API active; menu item commented in frontend.	Partially Connected
F054	Solar Store (EPC)	Order Status Tracking	Real-time order tracking portal showing order verification, warehouse picking, dispatch transporter details, and delivery milestones.	Fully Connected
F055	SolarShop India	Landing Page	Consumer retail landing page showcasing solar benefits and residential rooftop packages (currently using static mock content).	Frontend Only
F056	SolarShop India	Industries Showcase Page	Industry showcase page presenting tailored solar solutions for agriculture, commercial, industrial, and residential sectors.	Partially Connected
F057	BOSKIT	Distributor Onboarding	Multi-step onboarding wizard for BOS channel distributors featuring GST verification, KYC document uploads, and territory allocation.	Fully Connected

F058	BOSKIT	Distributor Plans Management (Admin)	Admin management module to configure distributor subscription tiers, security deposits, territory exclusivity rights, and credit policies.	Fully Connected
F059	BOSKIT	Distributor Dashboard	Operational dashboard for BOS distributors to manage local dealer networks, track bulk purchase orders, and monitor regional territory sales.	Fully Connected
F060	BOSKIT	Distributor Procurement Catalogue	Wholesale procurement catalog allowing distributors to order BOS components at master trade rates and configure downstream dealer margins.	Fully Connected
F061	BOSKIT	Custom BOS Kit Configurator	Interactive configurator for designing electrical BOS distribution kits tailored to specific inverter capacities and solar string requirements.	Partially Connected
F062	BOSKIT	Dealer Registration & Dashboard	Self-service portal for local electrical dealers to register under regional distributors, browse catalogs, place orders, and track shipments.	Fully Connected
F063	BOSKIT	BOSKIT Admin Overview & Reports	Executive BI dashboard aggregating channel distribution performance, dealer turnover, and inventory movement across regional BOS networks.	Fully Connected
F064	BOSKIT	BOSKIT MOQ, GST, Channel Settings	Platform settings interface managing BOS minimum order quantities, HSN GST tax rules, and channel discount matrices.	Fully Connected
F065	BOSKIT	FranchisePlansAdminPage	Dedicated admin interface for managing franchise packages, perks, territory exclusivity clauses, and fee structures (currently a placeholder stub).	Not Started
F066	Payments & Accounts	Offline Payment & Receipt Upload System	Central financial transaction engine handling offline bank transfer slip uploads, UTR number capture, Cloudinary document storage, and accounts verification.	Fully Connected
F067	Payments & Accounts	Franchise Earning & Payout Mgmt (Admin)	Financial management panel for Accounts team to review franchisee earning withdrawal requests, process manual bank payouts via NEFT/RTGS, and log UTR numbers.	Fully Connected
F068	Payments & Accounts	Accounts Panel (Solar Shop Finance)	Financial management portal tracking franchise plan fee receipts, EPC direct transactions, and commission disbursements.	Fully Connected
F069	Payments & Accounts	Purchase Orders (Accounts)	Complete purchase order lifecycle tracking supplier invoices, bank payment slips, tax deductions, and milestone payments.	Fully Connected
F070	Payments & Accounts	Supplier Registry (Accounts)	Supplier financial registry maintaining vendor GSTIN, bank account details, MSME registration, and payment terms.	Fully Connected
F071	Warehouse	Material Inward (Stock Inward)	Warehouse inbound operations module logging receipt of manufacturer consignments, serial number scanning, and QA inspection reports.	Fully Connected
F072	Warehouse	Material Outward & Delivery	Warehouse dispatch engine managing picking lists, packing slips, transporter assignment, and e-Way bill generation.	Fully Connected
F073	Warehouse	Vehicle & Driver Management	Fleet logistics registry managing company transport vehicles, driver assignments, insurance renewals, and trip logs.	Fully Connected
F074	Warehouse	Inventory Transfer, Stock Adj, Replacement	Warehouse stock management tool facilitating inter-warehouse material transfers, damaged stock write-offs, and customer warranty replacements.	Partially Connected
F075	Warehouse	Repair Tickets	Maintenance service desk logging RMA tickets for defective solar inverters, panels, or meters requiring factory repair (backend route pending).	Frontend Only
F076	Operations	Operations Panel (Order Dispatch, Logistics)	Operations control center coordinating logistics milestones, courier tracking, and site delivery confirmations (backend API routes under-built).	Partially Connected
F077	Notifications	BDE Notifications	Automated notification feed alerting field agents about lead status changes, store inspection milestones, and target achievements.	Fully Connected
F078	Notifications	BOSKIT Notifications	Channel communication system alerting distributors and dealers about order dispatches, price revisions, and stock updates.	Fully Connected
F079	Notifications	Email (Nodemailer)	Automated backend email service dispatching KYC approval notices, invoice PDFs, and password reset links via SMTP.	Backend Only

F080	Notifications	SMS/WhatsApp (Twilio)	Instant messaging gateway sending one-time login passwords (OTP), delivery dispatch alerts, and critical account updates.	Backend Only
F081	Reports	BDE Performance & Conversion Funnel	Analytics dashboard evaluating field agent productivity, lead-to-onboarding conversion rates, and regional revenue contribution.	Fully Connected
F082	Reports	BOSKIT Cross-Platform Reports	Consolidated reporting engine providing executive visibility across distributor procurement, dealer turnover, and inventory velocity.	Fully Connected
F083	Reports	Franchise Analytics (Admin)	Business analytics evaluating franchise partner sales volume, average order values, and commission distribution.	Fully Connected
F084	Security	NoSQL Injection Prevention	Global security middleware (‘mongo.sanitize’) sanitizing all incoming payloads by stripping MongoDB operator keys (\$ and .).	Backend Only
F085	Security	Rate Limiting (Auth, GST, OTP)	API protection middleware throttling authentication attempts, OTP generation, and GST API calls to prevent brute-force attacks.	Backend Only
F086	Security	JWT with httpOnly Cookies	Secure authentication token architecture utilizing RS256 encryption with 15-minute access tokens and 2-day refresh cookies.	Backend Only
F087	Security	CORS Vulnerability (Over-permissive)	CORS origin callback in server bootstrap currently allows all origins unconditionally (requires immediate restriction).	Backend Only
F088	Security	Receipt Slip File Upload Validation	Multipart form validation middleware inspecting uploaded payment slips for valid image/PDF MIME types and file size bounds.	Backend Only
F089	Security	KYC & Payment Document Private Delivery	Document storage pipeline saving sensitive verification documents and payment slips to Cloudinary (requires enforcement of private signed URLs).	Backend Only
F090	Deployment	Docker Multi-Stage Build	Production containerization script creating optimized, minimal Node.js and static web server runtime containers.	Backend Only
F091	Deployment	Nginx Reverse Proxy	Reverse proxy routing HTTP/HTTPS traffic to internal port services with SSL termination and gzip compression.	Backend Only
F092	Deployment	Render.com Live Deployment	Cloud platform hosting running live instances of central backend and web portals on Render.com.	Fully Connected

7. Bug & Risk Register

ID	Module	Issue / Risk Description	Evidence Location	Severity
B001	Security	CORS allow-all override — index.js:45 returns callback(null,true) unconditionally, bypassing whitelist	backend/ solarkits-central-backend/src/ index.js:45	CRITICAL
B002	Security	Payment receipts and KYC documents via Cloudinary may use public delivery instead of private signed URLs	.env.example CLOUDINARY_KY C_UPLOAD_PRESET commented out	HIGH
B003	Solar Store (EPC)	'Request Order' menu item has no route mapping in Board.jsx — 404 on navigation	solar-store/src/Pages/Board.jsx — /request-order route not registered	MEDIUM
B004	Franchise Portal	PlansPortal route commented out in App.jsx — users cannot browse plans	customer-apps/ solarkits-reseller-portal/src/ App.jsx	MEDIUM
B005	BOSKIT Admin	FranchisePlansAdminPage.jsx is a 111-byte placeholder stub — not implemented	boskit/pages/ FranchisePlansAdminPage.jsx	MEDIUM
B006	Franchise Portal	LooseOrder.jsx is a 136-byte placeholder stub — feature not implemented	customer-apps/ solarkits-reseller-portal/src/ pages/LooseOrder.jsx	LOW
B007	Solar Store (EPC)	SolarBosKit component exists and backend APIs present, but menu item commented out	solar-store/src/Pages/Board.jsx — Solar BOS Kit commented	LOW
B008	Warehouse	Repair Tickets UI exists but no backend route/controller found	warehouse/RepairTickets.jsx vs warehouse.route.js	MEDIUM
B009	Operations	Operations portal has full UI but backend has only 1 route file with minimal endpoints	operation-management/routes/ root.route.js	HIGH
B010	SolarShop India	SolarShop India landing page is static — no live product or API data connected	solarkits-solarshop-india/src/ App.jsx — no API calls at landing	MEDIUM
B011	Admin Panel	Admin dashboard Home.jsx is a 376-byte placeholder — no real statistics	internal-admin-portal Home.jsx	MEDIUM
B012	Documentation	Legacy payment documentation still references online Razorpay gateway instead of active Offline Receipt Upload workflow	payment_documentation.md	MEDIUM

8. Security & Data Privacy Analysis

Level	Security Area	Audit Finding & Evidence
CRITICAL	CORS Allow-All Override	index.js:45 — The CORS callback always returns callback(null, true) unconditionally. The origin whitelist above it is completely bypassed. Every HTTP origin is permitted.
HIGH	Payment Receipt & KYC Private Delivery Not Confirmed	.env.example — CLOUDINARY_KYC_UPLOAD_PRESET is commented out. Bank transfer receipt slips, PAN, Aadhaar, and GST documents may be publicly downloadable via Cloudinary URLs.
OK	Payment Receipt Slip File Validation	kycDocUpload / multer validation applied on receipt uploads. Validates uploaded slip file types and prevents unauthorized execution.
OK	NoSQL Injection Prevention	mongo.sanitize middleware applied globally in index.js:57. Strips \$ and . operators from all incoming request bodies, query strings, and params.
OK	Rate Limiting (Auth, GST, OTP)	authRateLimiter, gstRateLimiter, and otpRateLimiter are applied on sensitive endpoints.
OK	JWT with httpOnly Cookies	RSA key pair used for JWT signing. Access tokens expire in 15 minutes; Refresh tokens expire in 2 days. httpOnly cookie flag prevents JavaScript XSS theft.
OK	bcrypt Password Hashing	bcrypt v6 listed in backend package.json. Standard secure password hashing.
OK	B2B Price Isolation (Tenant Separation)	EPC catalogue API (epc-catalogue endpoint) never exposes franchise purchase price or margin limits.
NOTE	Trust Proxy Configuration	app.set('trust proxy', 1) is set. Ensure Nginx passes correct X-Forwarded-For headers.
NOTE	No Automated Test Suite	No Jest, Mocha, or Supertest dependencies found. No regression protection for receipt upload/auth flows.

9. Module Completion Estimates

Module / Component	Completion	Assessment	Key Architectural Notes & Gaps
Backend API (all modules)	82%	Well-developed	Customers module stub; Operations severely limited
Internal Admin Portal	70%	Good progress	Admin home.jsx is 376-byte placeholder
BDE Module (Admin + Field Portal)	78%	Good progress	Fully featured; minor UI stubs
Franchise Portal	75%	Good progress	PlansPortal route commented; LooseOrder stub; receipt upload verified
Solar Store (Direct EPC)	72%	Good progress	Request Order not mapped; BOS Kit menu commented
SolarShop India Marketplace	25%	Early stage	Static landing page; no live API connected
BOSKIT B2B Platform	68%	Good progress	FranchisePlansAdminPage is 111-byte stub
Offline Payments & Accounts Engine	85%	Well-developed	Offline receipt upload coded; requires private Cloudinary enforcement
Warehouse Module	70%	Good progress	Repair Tickets frontend only; Transfer partial
Operations Module	30%	Early stage	Full UI present; 1 backend route file only
Reports & Analytics	60%	Moderate	BDE funnel + BOSKIT + franchise reports coded
Security & Infrastructure	70%	Good progress	CORS bypass + receipt/KYC delivery are open issues
OVERALL PLATFORM	77.0%	Substantial progress	Security fixes + Operations / SolarShop India completion required

10. Offline Payment & Receipt Verification Workflow

Architecture: 100% Offline Bank Transfer with Receipt Slip Upload and Admin/Accounts Verification. The system does NOT use an online payment gateway. All payments follow the verified 4-step workflow below:

Step 1: Offline Transfer: Customer / Franchise partner executes NEFT / RTGS / IMPS / Cheque payment to Company Bank Account.

Step 2: Receipt Upload: User uploads Payment Slip + UTR + Bank Name via POST /api/india/v1/reseller/fee-payment/upload-receipt. Status becomes 'receipt_uploaded'.

Step 3: Admin Review: Admin / Accounts team views receipt image and validates UTR number against bank statement.

Step 4: Approval & Activation: Admin approves payment (status becomes 'verified'). Operational dashboard unlocks and Franchise Earning is logged.

Payment / Accounts Feature	API Route / Controller Endpoint	Integration Status
Manual Fee Payment Receipt Upload	POST /api/india/v1/reseller/fee-payment/upload-receipt	Fully Connected
Franchise Fee Payment Status Check	GET /api/india/v1/reseller/auth/me (fee_payment_status)	Fully Connected
Admin Payment Receipt Verification	PUT /admin-api/reseller-mgmt/verify-receipt/:id	Fully Connected
Franchise Activation on Receipt Approval	reseller.activation.service.js (UTR check)	Fully Connected
Franchise Earning Auto-Credit	Commission service — auto on EPC order	Fully Connected
Withdrawal Request (Franchise Partner)	POST /api/india/v1/reseller/wallet/withdraw	Fully Connected
Franchise Earning Payout with UTR (Admin)	PUT /admin-api/reseller-mgmt/wallet/payouts/process/:id	Fully Connected
Commission & Earning Settlement (Admin)	POST /admin-api/reseller-mgmt/wallet/settle-commission	Fully Connected
Export Payouts CSV (Admin)	GET /admin-api/reseller-mgmt/wallet/export-payouts	Fully Connected
Accounts Dashboard Financial Stats	/account-api/accounts/solar-shop/dashboard-stats	Fully Connected
Franchise Plan Payments (Accounts)	/account-api/accounts/solar-shop/franchise-plans	Fully Connected
Franchise Earning & Commission Tracking	/account-api/accounts/solar-shop/franchise-commissions	Fully Connected
Direct EPC Transactions (Accounts)	/account-api/accounts/solar-shop/direct-epc-transactions	Fully Connected
Purchase Orders & Receipt Logs (Accounts)	/account-api/accounts/purchase-orders/*	Fully Connected

11. Recommended Development Roadmap

Priority	Module	Action Required	Business Impact	Effort
P0	Security	Fix CORS allow-all override in index.js:45	All API endpoints accept any origin — data exposure risk	Small
P0	Security	Implement Cloudinary private signed URL delivery for KYC & Payment Receipt documents	Payment slips, PAN/Aadhaar/GST docs may be publicly accessible	Medium
P1	Solar Store	Implement /request-order route and page in Board.jsx	Menu item leads to broken navigation for EPC users	Medium
P1	Operations	Build out Operations Portal backend API routes	Order dispatch and logistics are non-functional	Large
P1	Warehouse	Implement backend routes for Repair Tickets, Inventory Transfer, Stock Adjustment	Warehouse staff cannot manage repairs or transfers	Medium
P1	Franchise Portal	Re-enable PlansPortal route in App.jsx	Franchise partners cannot browse plans without knowing direct URL	Small
P1	Admin Panel	Build complete Customers management module (573-byte stub currently)	Customer management entirely absent from admin	Large
P2	SolarShop India	Connect landing page to CMS/product API	Marketplace shows static data, not live products	Medium
P2	BOSKIT	Implement FranchisePlansAdminPage.jsx (currently 111-byte stub)	Franchise plan management absent from BOSKIT admin	Medium
P2	Franchise Portal	Implement LooseOrder.jsx functionality	Loose order feature unavailable to franchise partners	Medium
P2	Solar Store	Re-enable Solar BOS Kit menu item	BOS kit purchasing inaccessible from Solar Store navigation	Small
P3	Admin Panel	Build real Admin dashboard Home.jsx with statistics	Admin portal home page is a blank placeholder	Medium
P3	Documentation	Update payment documentation to formalize Offline Payment & Receipt Verification Workflow	Prevents confusion with legacy gateway drafts	Small
P3	Documentation	Add Swagger/OpenAPI API documentation	Developer onboarding and QA testing difficult without docs	Large
P3	Testing	Implement automated end-to-end test suite (Jest/Supertest)	No regression protection for payment receipt upload and auth flows	Extra Large

12. Conclusion & Priority Actions

SolarKits v2.0 demonstrates sophisticated enterprise-grade architecture with domain-driven module design, a 7-database MongoDB structure, offline payment receipt verification workflows, and robust role-based access control across 12 distinct user types.

The code-audit-based completion estimate of **77.0%** reflects substantial progress across all modules. Key strengths include the complete BDE lifecycle pipeline, comprehensive franchise API coverage, offline receipt upload & verification engine, and the BOSKIT distributor-dealer ecosystem.

Immediate Priority Actions:

- [P0 — CRITICAL] Fix CORS allow-all override in backend/src/index.js:45
- [P0 — CRITICAL] Enforce Cloudinary private signed URLs for payment receipts & KYC documents
- [P1 — HIGH] Build Operations Portal backend API routes (order dispatch, logistics)
- [P1 — HIGH] Complete Warehouse backend routes: Repair Tickets, Inventory Transfer, Stock Adjustment
- [P1 — HIGH] Re-enable PlansPortal route in Franchise Portal App.jsx
- [P1 — MEDIUM] Map /request-order route in Solar Store Board.jsx
- [P2 — MEDIUM] Connect SolarShop India landing page to live product/CMS API
- [P2 — MEDIUM] Implement BOSKIT FranchisePlansAdminPage.jsx and LooseOrder.jsx features
- [P3 — LOW] Build real Admin dashboard Home.jsx with real-time platform statistics
- [P3 — LOW] Add Swagger/OpenAPI documentation for all 9 backend modules

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